

- b) To be a citizen of this country, it is sufficient that you were born in the United States.
- Ans: If you were born in the United States, then you must be a citizen of this country.
- c) If you keep your textbook, it will be a useful reference ~~for~~ in your future courses.
- Ans: If you keep your textbook, then it will be a useful reference in your future courses.
- d) The Red Wings will win the Stanley Cup if their goalie plays well.
- Ans: If their goalie plays well, then the Red Wings will win the Stanley Cup.
- e) That you get the job implies ^{that} you had the best credentials.
- Ans: If you get the job, then you had the best credentials.
- f) The beach erodes whenever there is a storm.
- Ans: If there is a storm, then the beach erodes.
- g) It is necessary to have a valid password to log on to the server.
- Ans: If someone wants to log on to the server, then that person must have a valid password.
- h) You will reach the summit unless you begin your climb too late.
- Ans: If you don't begin your climb too late, then you will reach the summit.
- i) You will get a free ice cream cone, provided that you are among the first 100 customers tomorrow.
- Ans: If you are among the first 100 customers tomorrow, then you will get a free ice-cream cone.

23. Construct a truth table for each of these compound propositions.

	P	$\neg P$	$P \wedge \neg P$	$P \vee \neg P$
a) $P \wedge \neg P$	T	F	F	T
b) $P \vee \neg P$	F	T	F	T